

Data Warehousing Project 2

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Assumptions

1. The pipeline is parallelized wherever possible, with data quality and correctness taking precedence over performance. While facts and dimensions could be processed in parallel by skipping the joins, correctness is ensured by performing these joins, even at the cost of reduced parallelization. As a result, the ETL follows three main sequential control flows: data staging, dimensions, and facts.
2. Master CSV files from Airbus, Boeing, and other sources are thoroughly vetted and reliable. As such, additional data validation checks are unnecessary, ensuring the ETL process focuses on transformation and integration without redundant verification steps.
3. Modularity, order, maintainability, separation of responsibilities, and soundness are prioritized to ensure a robust and scalable ETL design. To uphold these principles, some minor redundancy, such as multiple ingestions of the same data (e.g., dimensions), is accepted. This trade-off avoids compromising the overall design for negligible improvements in execution time.
4. A staging area is integral to the ETL process, offering benefits such as reproducibility, enhanced debugging capabilities, and enforcement of quality checks and business rules. This intermediate layer introduces a critical data breakpoint, facilitating both quality assurance and modularity. It ensures that only high-quality, validated data is passed to downstream dimension and fact data flows, resulting in a superior design framework.
5. The pipeline has been programmed to run parallelly; however, the current Talend version doesn't support concurrency. Therefore, the efficiency and parallelism must be assumed.
6. In order to gain recoverability, we have not only added the data staging job with its consequent database updates but also generation of intermediate CSV files after important operations like joins to be able to assess unexpected behavior.